

Rank fragility is driven by foreground structure, not grid-rate noise

Screening envelope				Wide envelope			
	latent r=0.0	latent r=0.5	latent r=1.0		latent r=0.0	latent r=0.5	latent r=1.0
grid only	100.0%	100.0%	100.0%		100.0%	100.0%	100.0%
use phase only	98.3%	99.9%	100.0%		79.5%	91.3%	100.0%
embodied only	86.8%	96.2%	100.0%		63.4%	75.6%	100.0%
service equivalence only	96.0%	99.5%	100.0%		73.9%	86.5%	100.0%
all four blocks	76.2%	87.8%	100.0%		55.0%	64.5%	100.0%

Two-phase first-rank frequency in declared stress designs

Each cell uses 20,000 seeded draws with the same triangular marginals.

The latent Gaussian-copula correlation changes dependence among technology-specific multipliers; it is not an empirical estimate.

Grid-only perturbation is shared across technologies. Common-mode foreground errors preserve more of the released ordering than independent errors.